A Hundred Years of Business Cycles and the Phillips Curve

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Motivation

- **Phillips curve** declared **dead**
  - Killed by policy or by luck
  - Non-linear, unstable

- **Aggregate evidence** is **inconclusive**
  - No forecasting performance (Stock and Watson, 2008)
  - Weak identification (Mavroeidis, Plagborg-Møller and Stock, 2014)

- **Cross-sectional evidence**: **flat**
  - Estimated without using aggregate variation (Hazell et al., 2022)

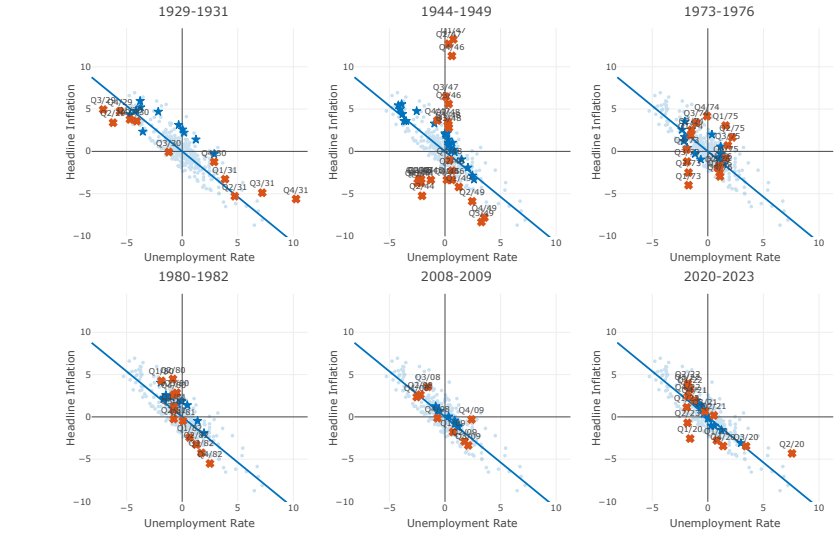
This Paper

- **Intuition**: raw data are **trend + cycle + noise**

- **Reduced-form TC model** consistent with a **NK economy** with stochastic trends
  - Idiosyncratic trends + one **common trend inflation**
  - One **common business cycle component**
  - Noise absorbed by idiosyncratic cyclical components

- **Long view**: seven quarterly US series from **1900**
  - Output, unemployment, headline and core CPI inflation, SPF and UoM expectations, policy rate

Trends and Noise Hide the Cycle



Raw data vs **cycles**: once trend and noise are removed, **steep Phillips curve** at business cycle frequency

Results

- A **single common cycle** drives US fluctuations, and it looks like a **demand cycle**
  - Matches NBER dates and the CBO and Greenbook output gaps
- The **policy rate** is mainly driven by the common cycle: **77%** of its cyclical variation
  - Idiosyncratic component small, mainly during the **Volcker disinflation**
- **Reduced-form Phillips curve alive and well**: -1.08 over the century
  - -0.40 in the raw data
  - -0.77 over **1946–84**, -0.86 over **1985–2019**
  - Trend inflation is key: holding it constant makes the curve unstable
- **Okun's law** is **at least as stable**: slope -0.417
  - -0.33 over **1946–84**, -0.52 over **1985–2019**
  - Ball, Leigh and Loungani (2017): -0.421

A Stylised NK Model with Trends

Trend Decomposition

Denote by $\widehat{x}_t$ the **cyclical component** of a variable x_t, and by τ_t^x its **trend**

- Output: $y_t = \tau_t^y + \widehat{y}_t$, **potential output** τ_t^y is I(2) with a stochastic growth rate
- Unemployment: $u_t = \tau_t^u + \widehat{u}_t$, the **NAIRU** τ_t^u is a driftless random walk
  - Cycle connected to output via **Okun's law**: $\widehat{u}_t = -\delta_u(L)\,\widehat{y}_t$
- Inflation: $\pi_t = \tau_t^\pi + \widehat{\pi}_t$, trend inflation τ_t^π is a driftless random walk
  - **Inflation expectations**: $E_t\pi_{t+1} = \tau_t^\pi + E_t\widehat{\pi}_{t+1}$, by the martingale property
  - Iterating forward: $\tau_t^\pi = \lim_{h\to\infty} E_t\pi_{t+h}$
- **Policy rate**: $i_t = \tau_t^r + \tau_t^\pi + \widehat{i}_t$, neutral real rate + trend inflation + cycle

Three-Equation Model and Solution

- **Three-equation model** in the **cyclical components**

$$\widehat{\pi}_t = \beta E_t \widehat{\pi}_{t+1} + \kappa\, \widehat{y}_t + u_t^s$$
$$\widehat{y}_t = \alpha \widehat{y}_{t-1} + (1-\alpha) E_t \widehat{y}_{t+1} - \sigma\left(\widehat{\imath}_t - E_t \widehat{\pi}_{t+1} - u_t^d\right)$$
$$\widehat{\imath}_t = E_t \widehat{\pi}_{t+1} + \theta_\pi \widehat{\pi}_t + \theta_d u_t^d + u_t^{mp}$$

- **Solution**: a **single cyclical state**, $\widehat{y}_t$, determines every cyclical component
  - Unique bounded equilibrium for $\alpha > 0$, $\kappa > 0$ and $\theta_\pi > 0$
  - $\widehat{y}_{t-1}$ is the only endogenous predetermined variable

Trend-Cycle Representation

- **Observables** admit a **trend-cycle representation**

$$\begin{pmatrix} y_t \\ u_t \\ \pi_t \\ E_t\pi_{t+h} \\ i_t \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} \tau_t^y \\ \tau_t^u \\ \tau_t^\pi \\ \tau_t^r \end{pmatrix}}_{\text{trends}} + \underbrace{\begin{pmatrix} 1 \\ -\delta_u(L) \\ \boldsymbol{\Phi} \\ \gamma^h\Phi \\ \Phi(\gamma+\theta_\pi) \end{pmatrix} \widehat{y}_t}_{\text{common cycle}} + \underbrace{\begin{pmatrix} 0 \\ 0 \\ u_t^s \\ 0 \\ \theta_\pi u_t^s + \theta_d u_t^d + u_t^{mp} \end{pmatrix}}_{\text{wedges}}$$

- **Reduced-form Phillips curve** at business cycle frequency: $\widehat{\pi}_t = \boldsymbol{\Phi}\, \widehat{y}_t + u_t^s$
  - $\Phi \equiv \kappa/(1-\beta\gamma)$: reduced-form slope
  - $\gamma \in (0,1)$: persistence of the business cycle, the stable root of the rational expectations system

An Empirical Model of Trends and Cycles

Observation Equations

$$\begin{pmatrix} y_t \\ u_t \\ \pi_t \\ \pi_t^c \\ F_t^{spf}\pi_{t+4} \\ F_t^{uom}\pi_{t+4} \\ i_t \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \tau_t^y \\ \tau_t^u \\ \tau_t^\pi \\ \mu_t^{spf,\pi} \\ \mu_t^{uom,\pi} \\ \tau_t^r \end{pmatrix}}_{\text{Trends \& biases}} + \underbrace{\begin{pmatrix} 1 \\ \sum_{j=0}^{1}\gamma_{2,j}L^j \\ \sum_{j=0}^{1}\gamma_{3,j}L^j \\ \sum_{j=0}^{1}\gamma_{4,j}L^j \\ \sum_{j=0}^{2}\gamma_{5,j}L^j \\ \sum_{j=0}^{2}\gamma_{6,j}L^j \\ \sum_{j=0}^{2}\gamma_{7,j}L^j \end{pmatrix} \widehat{y}_t}_{\text{Common cycle}} + \underbrace{\begin{pmatrix} \psi_{1,t} \\ \psi_{2,t} \\ \psi_{3,t} \\ \psi_{4,t} \\ \psi_{5,t} \\ \psi_{6,t} \\ \psi_{7,t} \end{pmatrix}}_{\text{Idiosyncratic cycles}}$$

- **Trends**: zero-one loadings, **common trend inflation**, **survey biases**
- **Common business cycle component**: loading on output set to **one**, others free in sign
  - Lead-lag dynamics through the lag polynomials $\sum_j \gamma_{i,j} L^j$
- Idiosyncratic cycles: data-theory **wedges** (measurement error, misspecification)

Transition Equations

- **Potential output** is **I(2)**: trend growth is a driftless random walk

$$\tau_t^y = \tau_{t-1}^y + g_t^y, \qquad g_t^y = g_{t-1}^y + u_t^g$$

- **All other trends and biases** are **I(1)** driftless random walks

$$\tau_t^x = \tau_{t-1}^x + u_t^{\tau,x}, \qquad x \in \{u, \pi, r\}, \qquad \mu_t^{j,\pi} = \mu_{t-1}^{j,\pi} + u_t^{\mu,j}, \qquad j \in \{spf, uom\}$$

- **Cyclical components**: **ARMA(2,1)** à la Harvey (1985)
  - Simplest process with pseudo-cyclical behaviour: frequency λ and damping ρ estimated

$$\begin{pmatrix} \widehat{y}_t \\ \widehat{y}_t^* \end{pmatrix} = \rho \begin{pmatrix} \cos\lambda & \sin\lambda \\ -\sin\lambda & \cos\lambda \end{pmatrix} \begin{pmatrix} \widehat{y}_{t-1} \\ \widehat{y}_{t-1}^* \end{pmatrix} + \begin{pmatrix} v_t \\ v_t^* \end{pmatrix}$$

- Each stationary and non-stationary component is **orthogonal**

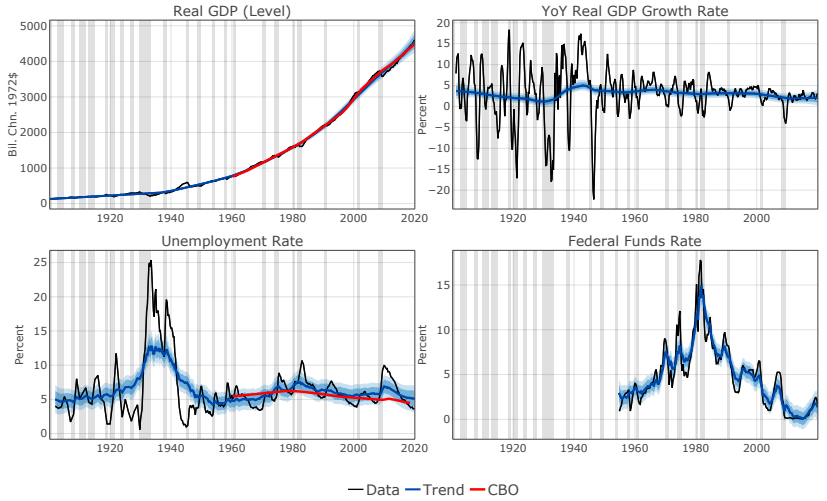
Data and Estimation

- **Bayesian estimation**: adaptive Metropolis-within-Gibbs and simulation smoother
- **Loose priors**: looser than Bianchi, Nicolò and Song (2026) and than the HP filter ▸ Priors

Variable	Transf.	Frequency	Period	Source
Real GDP	Log-levels	Q	1900–2019	Gordon (2007) NBER tables; FRED
Unemployment rate	Levels	A1901, Q1929	1901–2019	Haver Analytics
Inflation	YoY	A1914, Q1921	1914–2019	Haver Analytics
Core inflation	YoY	Q1957	1957–2019	FRED
Consumers' exp. inflation	Levels	Q1978	1978–2019	University of Michigan
SPF exp. inflation	Levels	S1946, Q1984	1946–2019	Livingston Survey; SPF
Short-term interest rate	Levels	Q1954	1954–2019	FRED

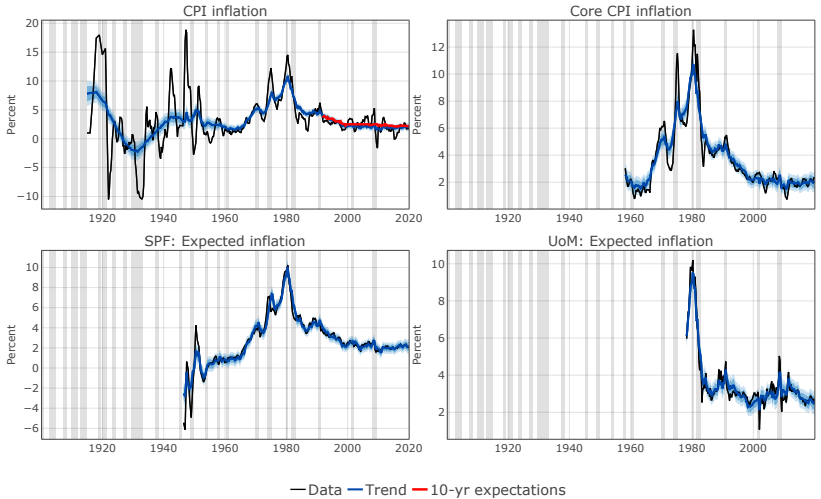
Trends and Cycles in the US Economy

Long Run Components (I/II)



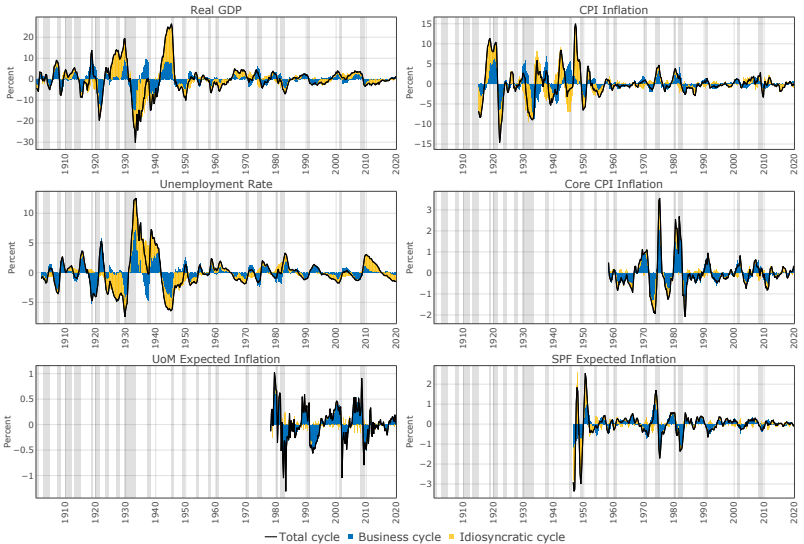
- **Potential output tracks the CBO**; trend growth slows from the 1970s

Long Run Components (II/II)

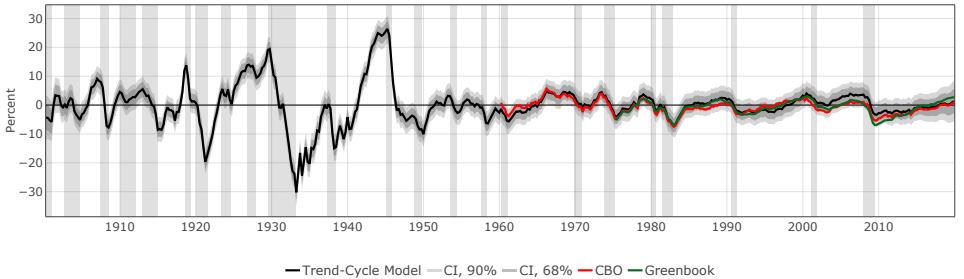


- **Trend inflation tracks the 10-year expectation**, which the model never sees

Common and Idiosyncratic Cycles

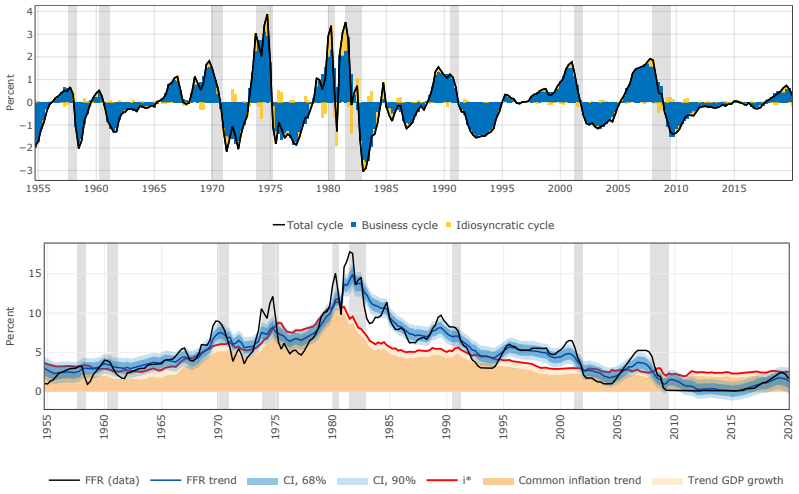


The Output Gap



- **Model-based output gap**: $y_t - \tau_t^y = \widehat{y}_t + \psi_{1,t}$
  - In percentage deviation from potential output
  - Sum of the common business cycle component and the idiosyncratic cycle of output
- Lines up with **NBER dates** and with the **CBO and Greenbook** gaps until the early 2000s

The Policy Rate: Cycle and Trend



- Policy-rate trend vs **natural nominal rate** $i_t^* \approx \tau_t^\pi + g_t$: the gap signals the stance

A Common Source of Business Cycle Fluctuations

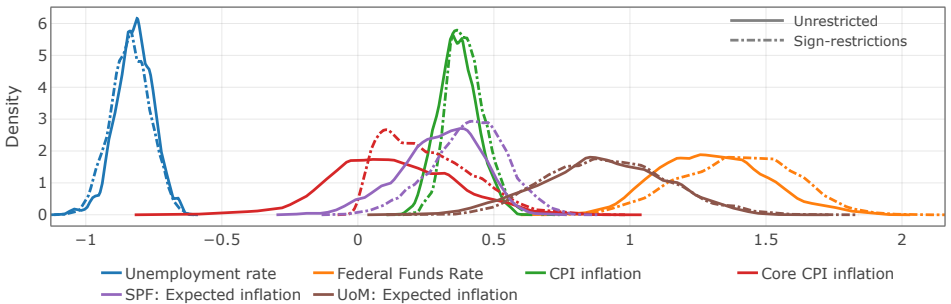
- The **common cycle** explains a **large share** of the cyclical variation of every series
  - Lower share for output, unemployment and CPI inflation: idiosyncratic movements before WWII
  - **77%** of the cyclical variation of the interest rate ▸ Policy rule
  - Robust when the model is fully re-estimated on subsamples ▸ Charts
  - Robust to correlated shocks with trend inflation, NAIRU and trend growth ▸ Charts
- **Not mechanical**: identified by comovement, not by maximising one variance (max share)

COMMON BUSINESS CYCLE SHARE OF THE CYCLICAL DECOMPOSITION (%)

y_t	u_t	π_t	π_t^c	$F_t^{spf}\pi_{t+4}$	$F_t^{uom}\pi_{t+4}$	i_t
41.8	46.1	45.2	61.1	50.5	65.3	**77.0**
[36.4, 47.7]	[40.3, 52.9]	[39.2, 50.0]	[43.8, 73.2]	[38.0, 62.0]	[50.0, 74.8]	[71.8, 81.0]

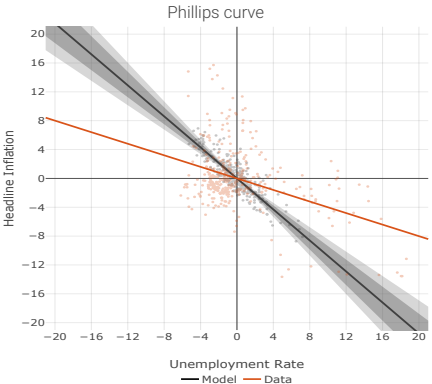
A Demand Cycle

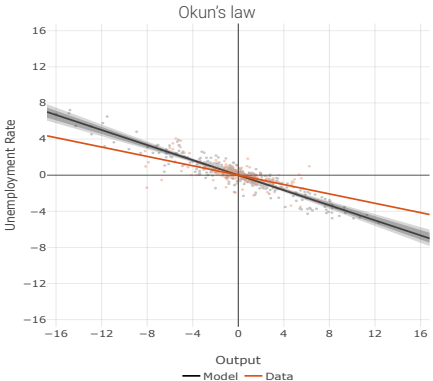
- The common cycle is a **demand cycle**: unemployment falls, prices and the policy rate rise
- Loadings are **not restricted** in sign: imposing sign restrictions only moves core inflation



Phillips Curve and Okun's Law

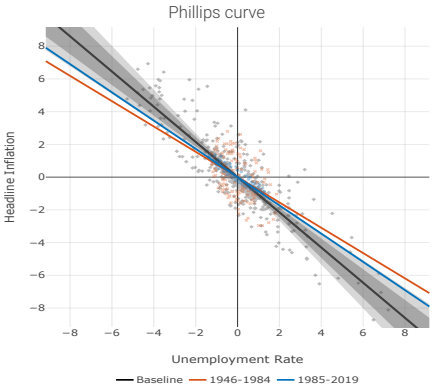
Phillips Curve and Okun's Law over 120 Years of Data

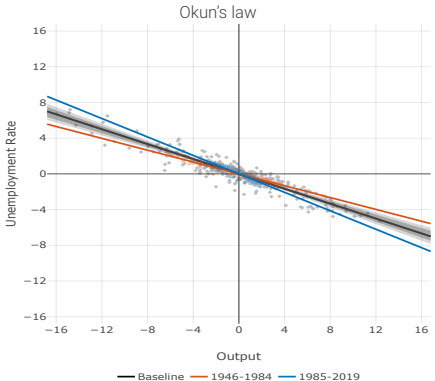




- **Regression coefficient** of the business cycle component of one variable on that of another
- **Phillips curve**: **−1.08** (−0.40 raw data)
- **Okun's law**: **−0.417** (−0.259 raw data)

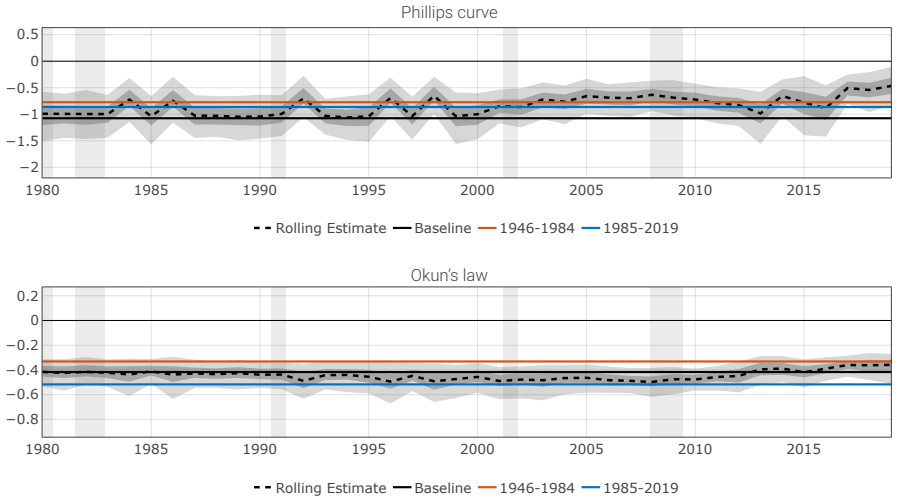
Stability Across Subsamples





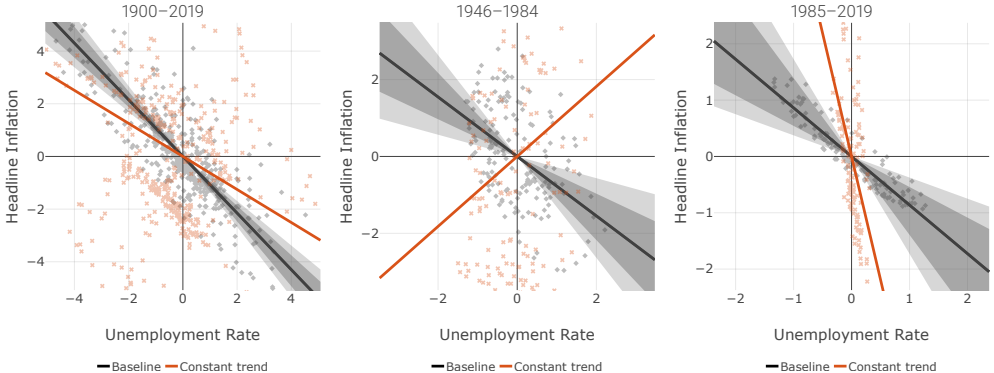
- Model **fully re-estimated** on each subsample, then regressions at business cycle frequency
- **Phillips curve**: **−0.77** over 1946–84, **−0.86** over 1985–2019
- **Okun's law**: **−0.33** over 1946–84, **−0.52** over 1985–2019

Rolling 80-Year Windows



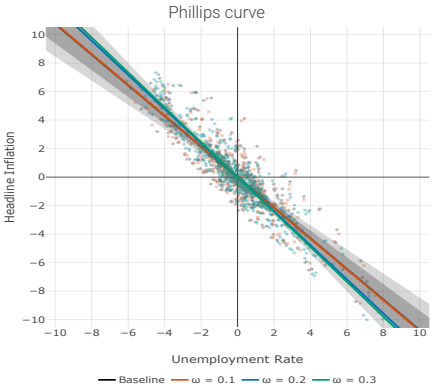
- Model fully re-estimated on **rolling 80-year windows**, ending from 1980 to 2019

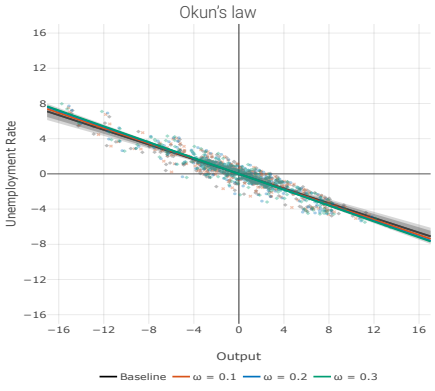
Phillips Curve: Trend Inflation Is the Key



- Constant **trend inflation**: the **Phillips curve** **flattens**, turns **positive**, then **almost vertical**

Robustness: Correlated Trend and Cycle Shocks





- Relax **orthogonality** of cycle and trend shocks (inflation, NAIRU, growth): prior scale ω
  - Prior sd of the cycle–trend correlation: 0.10, 0.18 and 0.25 for $\omega = 0.1, 0.2, 0.3$
- **Both slopes steepen** slightly: the baseline is, if anything, conservative

Conclusions

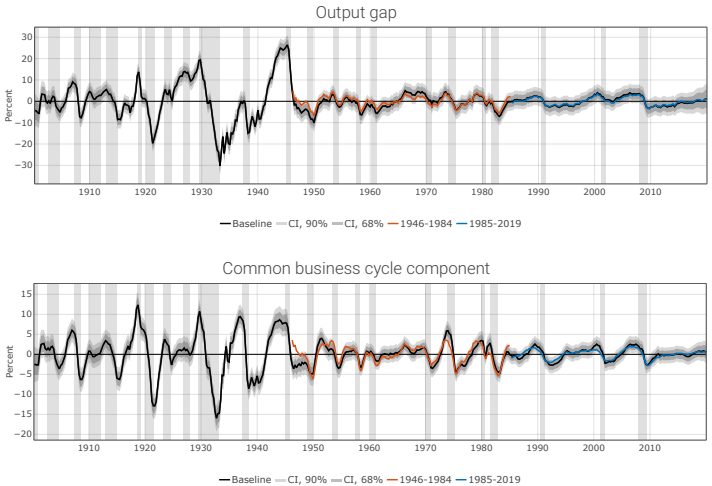
Conclusions

- A **single common cycle**, **demand-like**, drives a century of US fluctuations
  - Policy rate tightly connected to it: idiosyncratic movements mostly during the Volcker disinflation

- At business cycle frequency the **Phillips curve** is **steep and stable**
  - Robust across subsamples and across specifications
  - Controlling for trend inflation is essential: with a constant trend the curve disappears
  - **Okun's law** is at least as stable

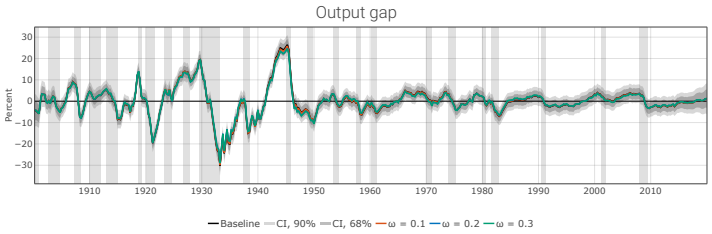
Appendix

Subsample Estimates



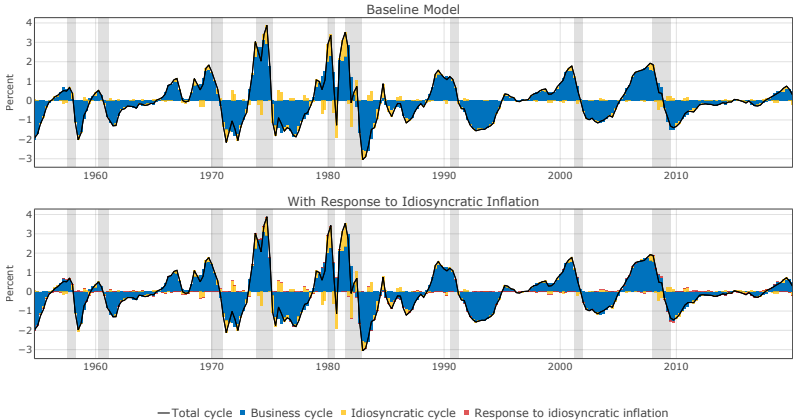
- Full sample against the two postwar subsamples, each fully re-estimated Back

Correlated Trend–Cycle Shocks



- Baseline (orthogonal) against $\omega = 0.1, 0.2, 0.3$ Back

Policy Rate and Idiosyncratic Inflation



- Policy rate also responds to the **idiosyncratic inflation cycle**: $\widehat{i}_t = \gamma(L)\widehat{y}_t + \theta_\pi \psi_{3,t} + \psi_{7,t}$
  - Taylor rule: tests whether the Fed responds to inflation off the demand cycle, e.g. supply shocks
- Response **small and short-lived**: the Fed largely looks through it Back

Prior Distributions

Name	Support	Density	Parameter 1	Parameter 2
γ (loadings)	$\mathbb{R}$	Normal	0	1000
τ_0 (initial trend levels)	$\mathbb{R}$	Normal	y_0	1000
σ^2 (shock variances)	$(0, \infty)$	Inverse-Gamma	3	1
σ_g^2 (trend-growth shock)	$(0, \infty)$	Inverse-Gamma	3	0.005
ρ (damping)	$[0.001,\ 0.97]$	Uniform	0.001	0.97
λ (frequency)	$[0.001,\ \pi]$	Uniform	0.001	π

- **Loose trend priors**: implied sd of the expected change in each trend Back
  - About 5.3% over 40 years for trend GDP growth, 6.6% over 20 years for the NAIRU
  - 6–8% over 20 years for trend real rate, trend inflation and survey biases (Bianchi et al.: 1%)
- **1946–84**: **training-sample prior** on the cycle, from a GDP, U, CPI model over 1900–45
  - Surveys sparse before 1984 (Livingston semi-annual, Michigan from 1978): excluded
  - Until core starts in 1957, trend inflation rests on headline CPI alone
  - CPI cycle distorted by price controls after WWII and during the Korean War